

PROJECT REPORT ON MODELLING AND SIMULATION OF SPOT ROBOT

Using Solidworks part modelling, assembly modelling and motion study tool

ABSTRACT

A Project inspired by the well-known SPOT ROBOT of the renowned robotics company: Boston Dynamics

By Aditya Ganta

National Institute of Technology
Tiruchirapally, Tamil Nadu
Student- BTech- Mechanical Engineering

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INTRODUCTION

SPOT is the dog inspired bot created by Boston Dynamics. Unlike typical bots that roll or glide SPOT can walk, climb and scramble through tricky spaces. This project draws inspiration from SPOT ROBOT. A working model of this project is also attached for perusal with this report.

Objective:

The objective of this project was the modelling and simulation of a SPOT ROBOT using SolidWorks part modelling, assembly modelling and motion study tool.

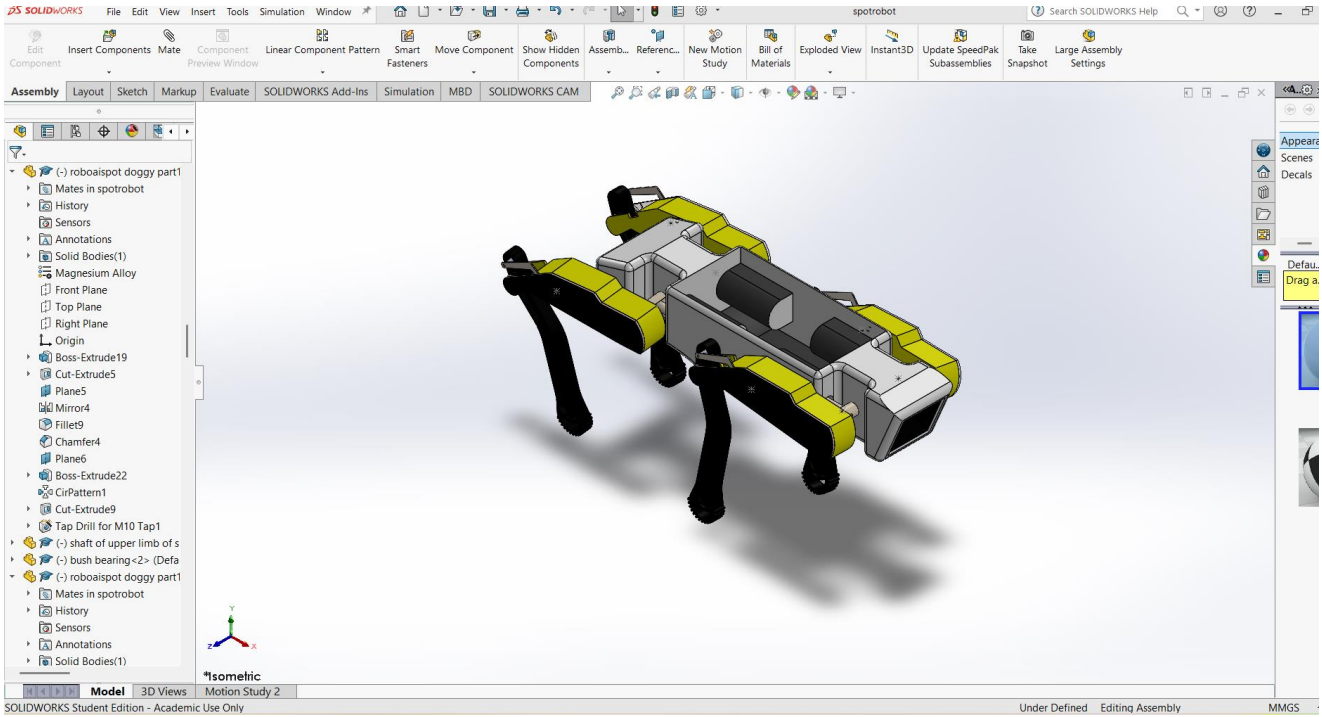
Background & Uses

The Spot Robot is a 4-legged quadruped robot dog that was designed for smooth mobility and agility in a vast plethora of environments. It uses its 4 legs like a dog in order to navigate through rough terrain, to climb stairs and to stand in a stable position even on an uneven surface. It can be operated manually by remote control or automatically using its perception and guidance systems in order to follow predefined roots. It is powered and run by multiple sensors and motors in each leg. Due its versatility it is used in a vast number of applications such as, industrial inspections in oil, gas and manufacturing industries, surveillance and security which involves monitoring hazardous areas, military and defense operations such as reconnaissance, remote operations and border patrol operations in areas where human monitoring can always be challenging. Apart from that Spot has also found its usefulness in disaster struck areas where it can enter sites which have been severely damaged like collapsed buildings and can detect survivors who are stuck using thermal detection cameras.





DESIGN METHODOLY



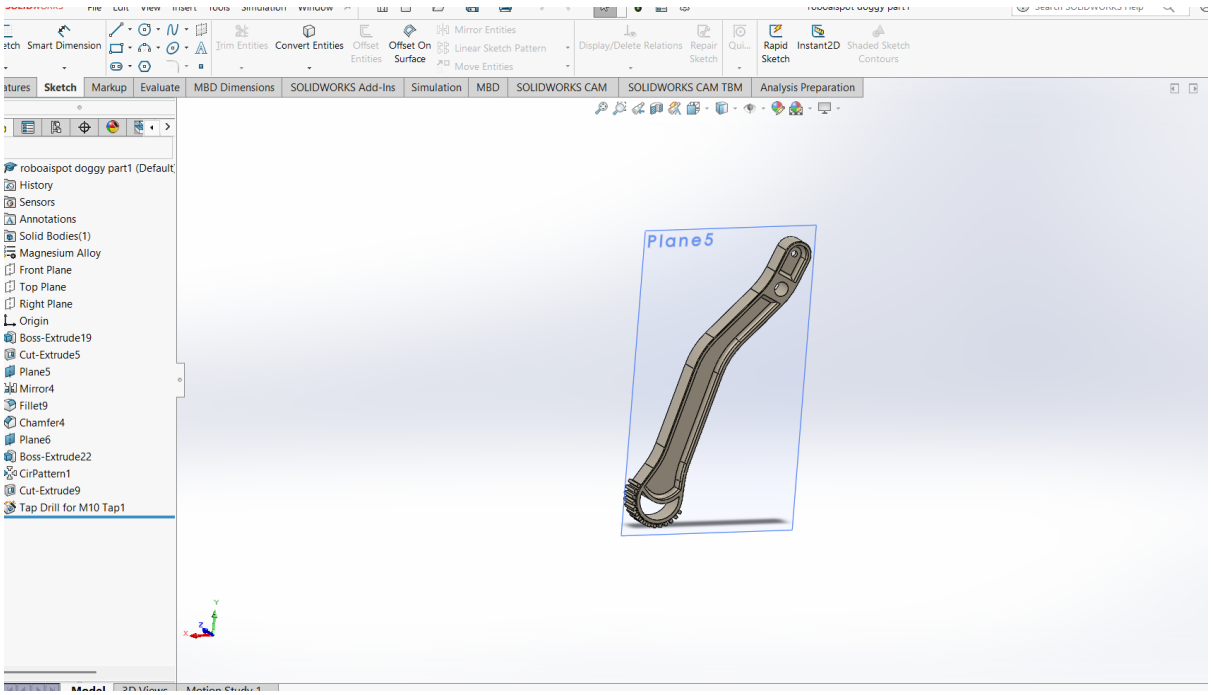
Final Model made on SolidWorks (Shown above)

As mentioned above in this report, this CAD model of the Boston Dynamics Spot Robot has been designed using the basic concepts of part modelling and assembly modelling. Therefore, I started with designing each of the parts individually. Given below is a brief description of some of the parts that I have modelled in SolidWorks.

INDIVIDUAL PARTS DESIGN

1) **Lower limb**:

This was designed by using the different sketch tools that are available on SolidWorks, and then extrusion followed by the extruded cut feature to make the holes that can be viewed in the part model. The lower limb consists of grips at the feet. These grips are used in order to provide stability to the robot dog especially during operations like climbing a flight of stairs. The material that was used in this case was Magnesium alloy with a density of 1070Kg/m^3(in addition to which a very thin film of protective coating can be added), because of its light weight, high strength, and cost effectiveness.

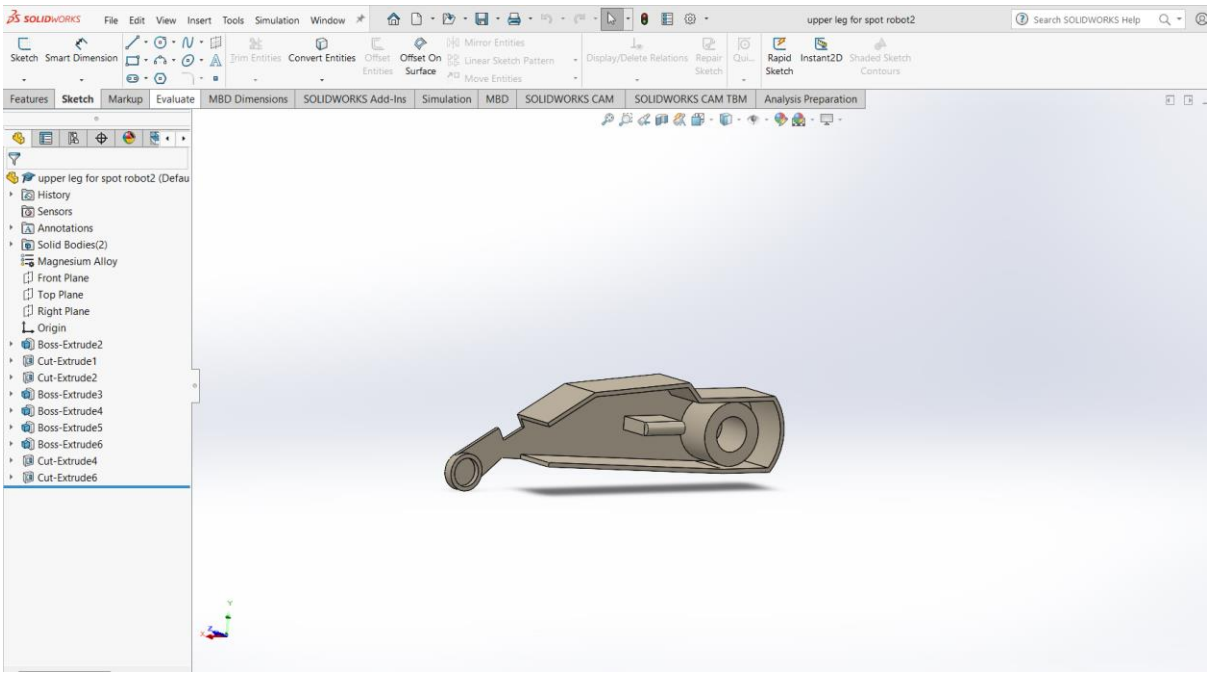


Lower Limb

2) **Upper limb:**

This was again designed by using the basic sketch tools in SolidWorks followed by some features like extrusion, extruded cut features. Using offset entities tool followed by extruded cut the upper leg was made hollow to accommodate the hub motors and the helical screw and nut mechanism which was used to make the legs move (this will be explained in more detail in the later half of this report). Magnesium alloy (1070Kg/m^3) has

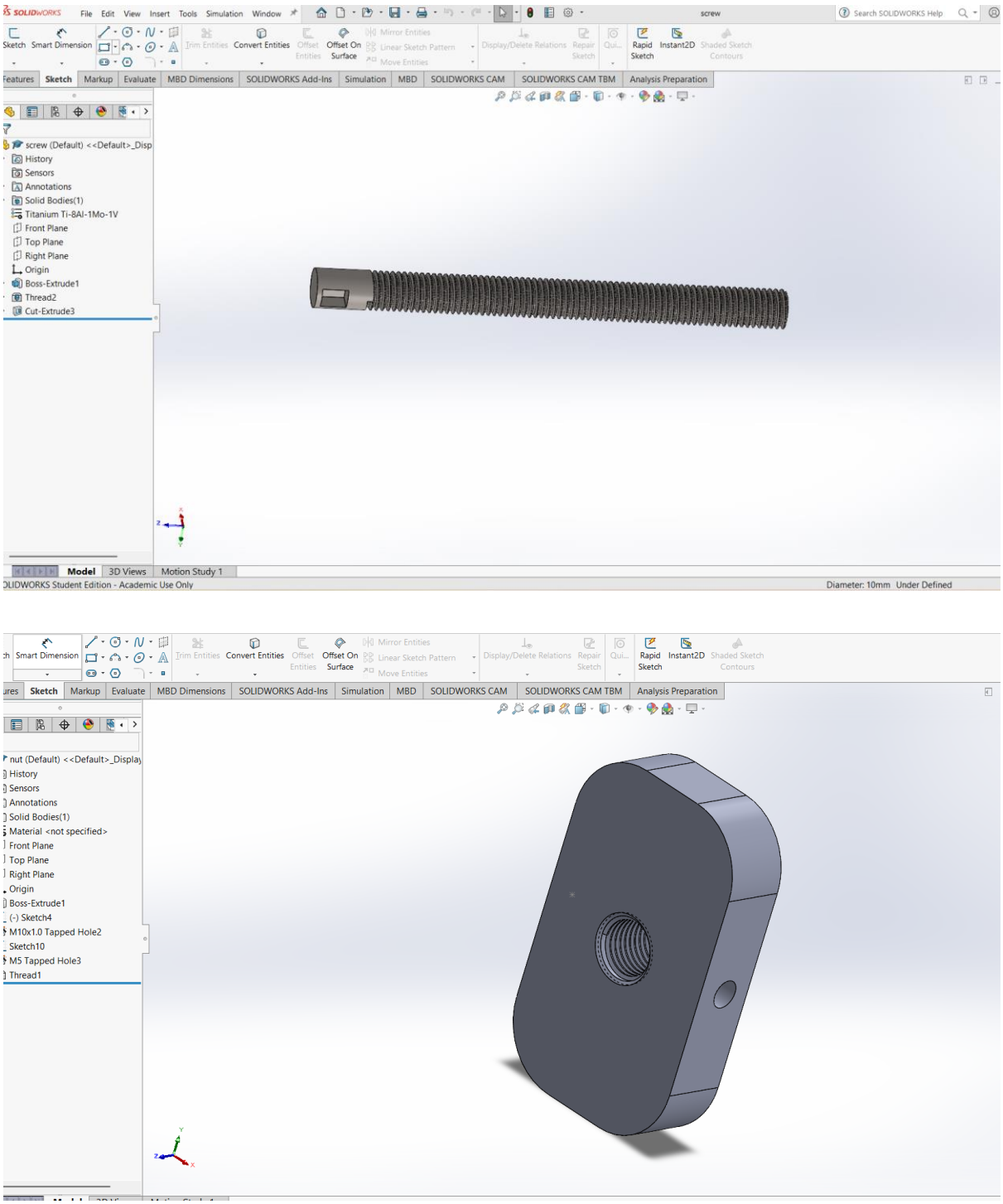
been used for the upper leg as it has a good heat dissipation rate and can dissipate excess heat that is generated from the motors and other electronic components while running.



Upper Limb

3) Screw and Nuts:

Screws and nuts are needed for constructing the helical mechanism within the upper limb of the Spot Robot. The screw was made by simple extruding a circular sketch and then applying threads on the cylindrical surface on the screw using the hole wizard command that is available in SolidWorks. The nut in this case has a unique shape of a rectangle due to design convenience which can be understood better from viewing the internal structure of the model. This nut has an M5 tapped hole on the sides which can be used to connect the screw nut mechanism to the lower limb. The material used in this case was the Titanium-Aluminum alloy that has a mass density of 4370 Kg/m^3.

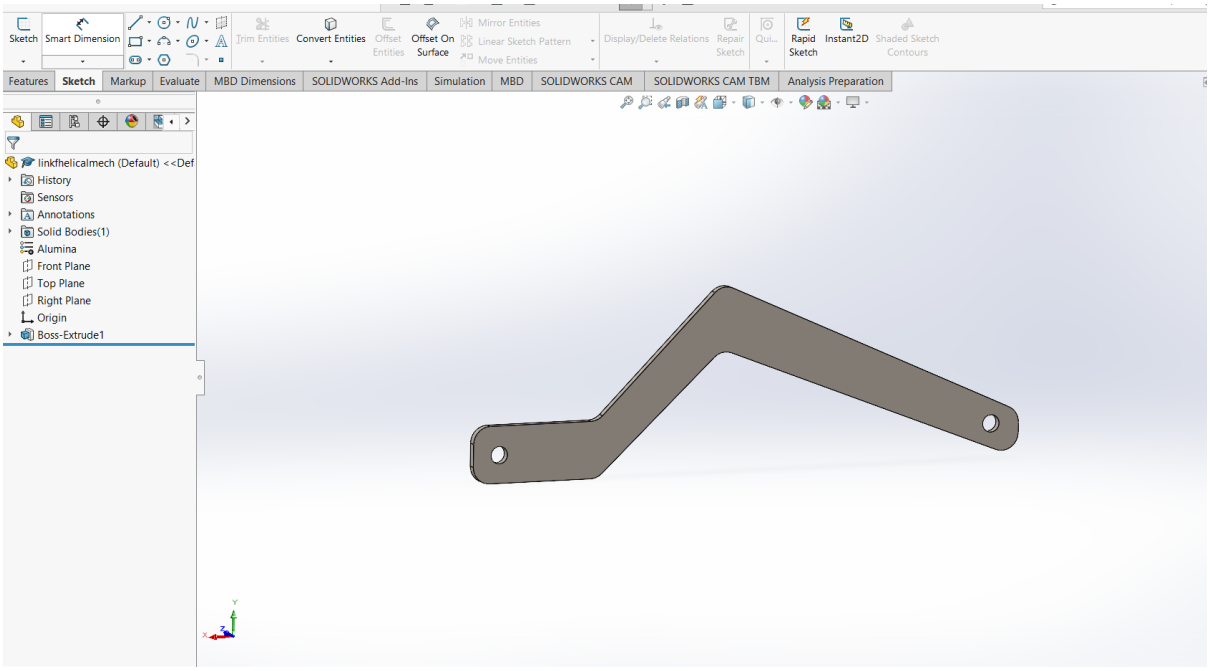


Screw & Nuts

4) <u>Connecting link:</u>

This was made to establish the connection between the upper and the lower limb. With this connection the actuation caused due to the motor is able to drive the lower limb. The material applied to this part was alumina alloy with a density of 3960Kg/m^3, which gives a good balance between

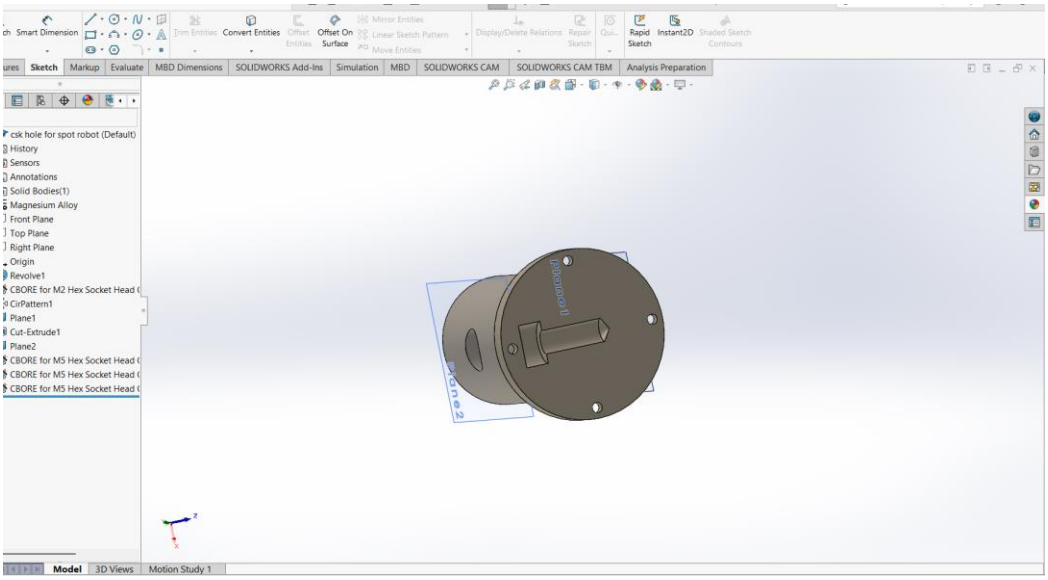
strength and weight. It was produced using simple sketch tools and 2 dimensional fillets followed by extrusion up to a very small distance of 2.5mm. The reason why it is called connecting link is that it establishes the final connection between the upper and lower limbs.



Connecting Link

5) Counter sunk key:

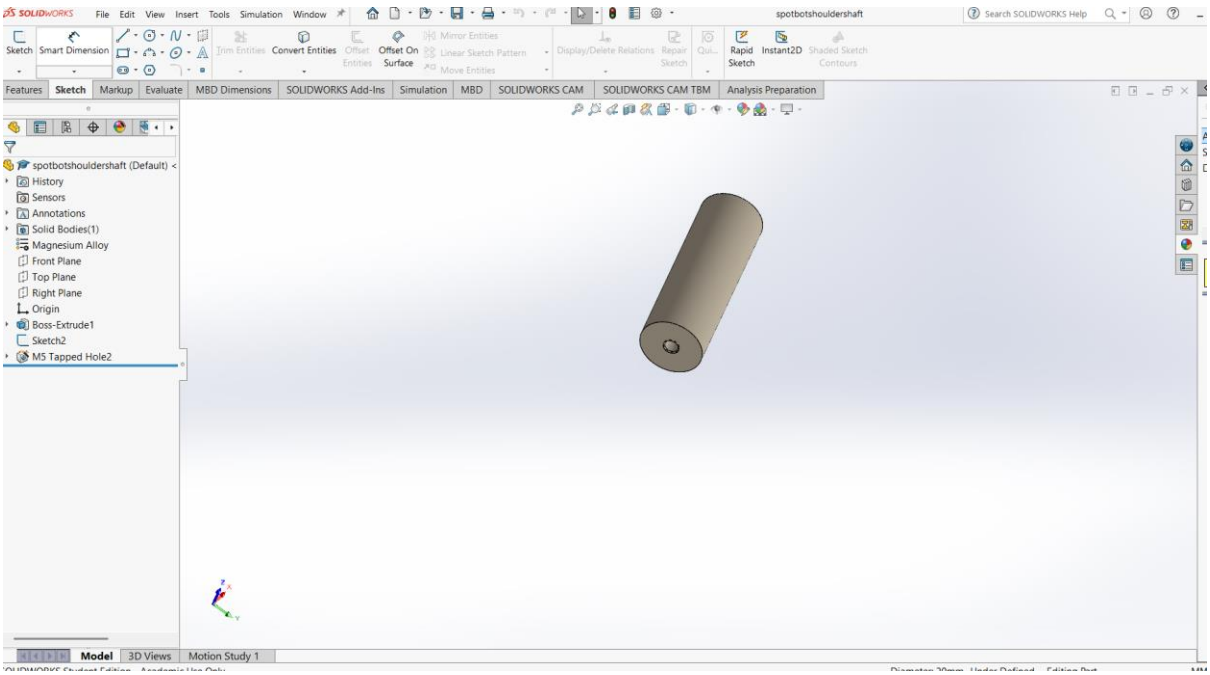
Used for accommodating the head of the fastener or bolt. It establishes a connection between the upper limb and the main body of the robot. It was made using extrusion, extruded cut and CSK hole feature in the hole wizard tab of SolidWorks. Magnesium alloy AZ31B was used in making this counter sunk key.



Counter Sunk Key

6) Shoulder shaft:

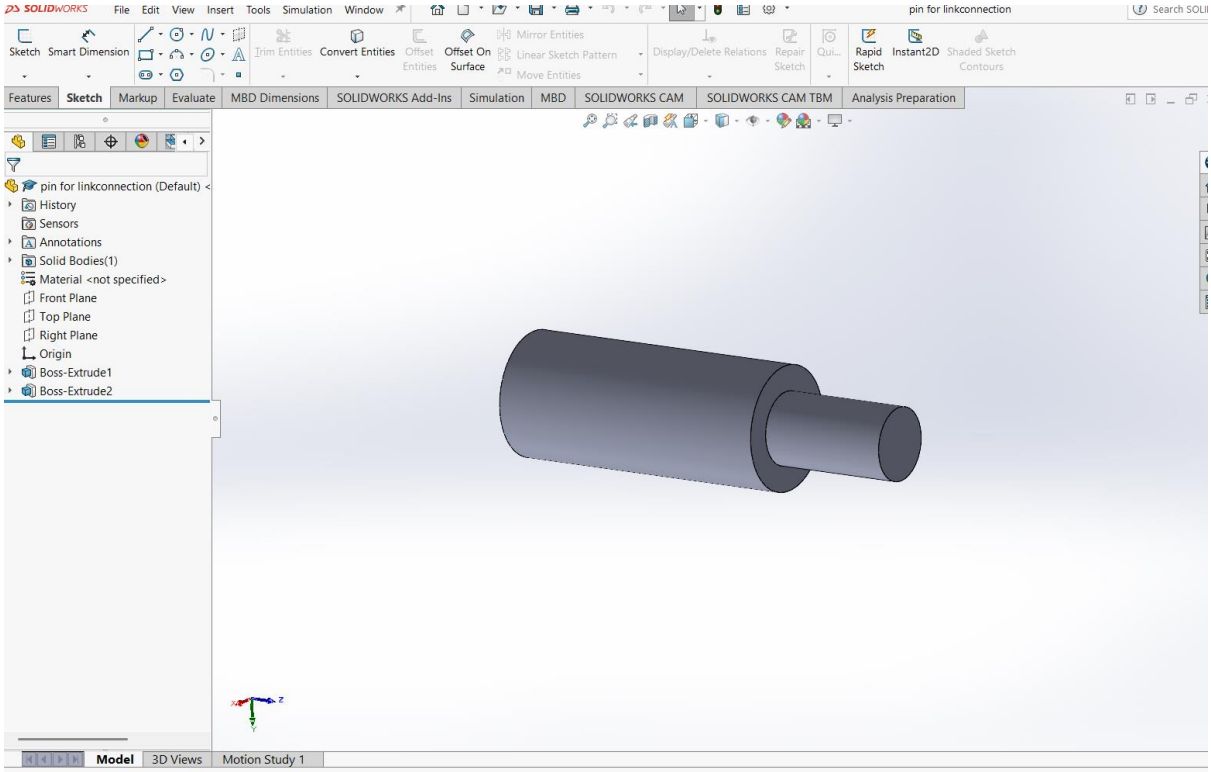
This is used for establishing a connection between the counter sunk and the main body. It was again made using the circular sketch tool and then simple extruding it to a desirable length. Finally using hole wizard an M5 tapped hole was made inside it. Material used here is also Magnesium alloy AZ31B.



Shoulder Shaft

7) Connecting pin:

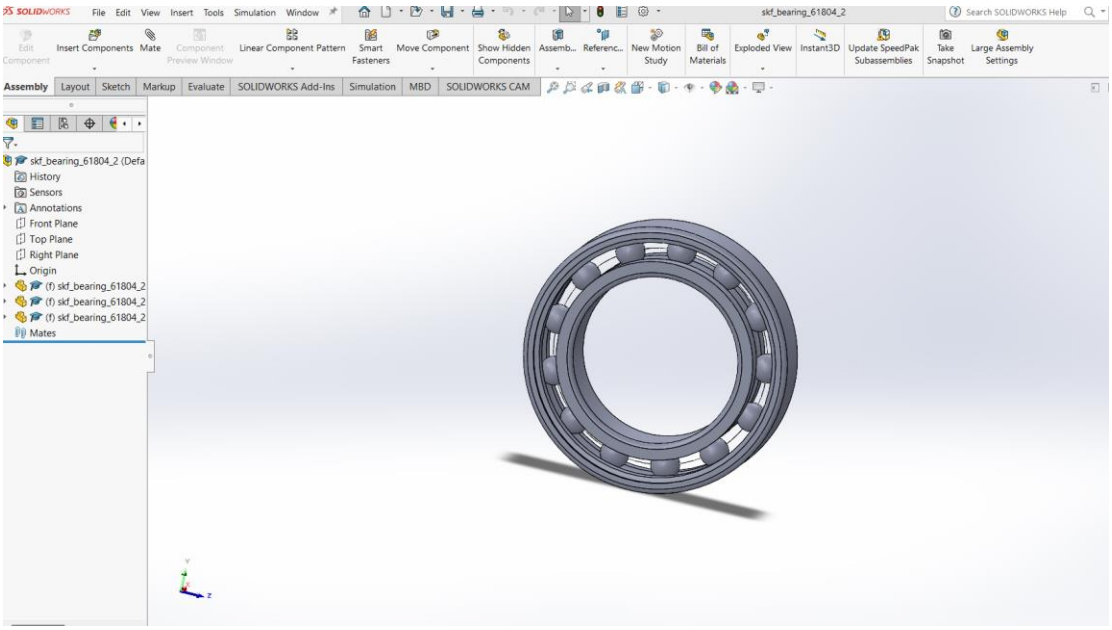
A pin that was designed in order to connect the connecting link to the lower limb. The 2 ends of this connecting pin are of two different diameters as shown.



Connecting Pin

8) Ball bearing:

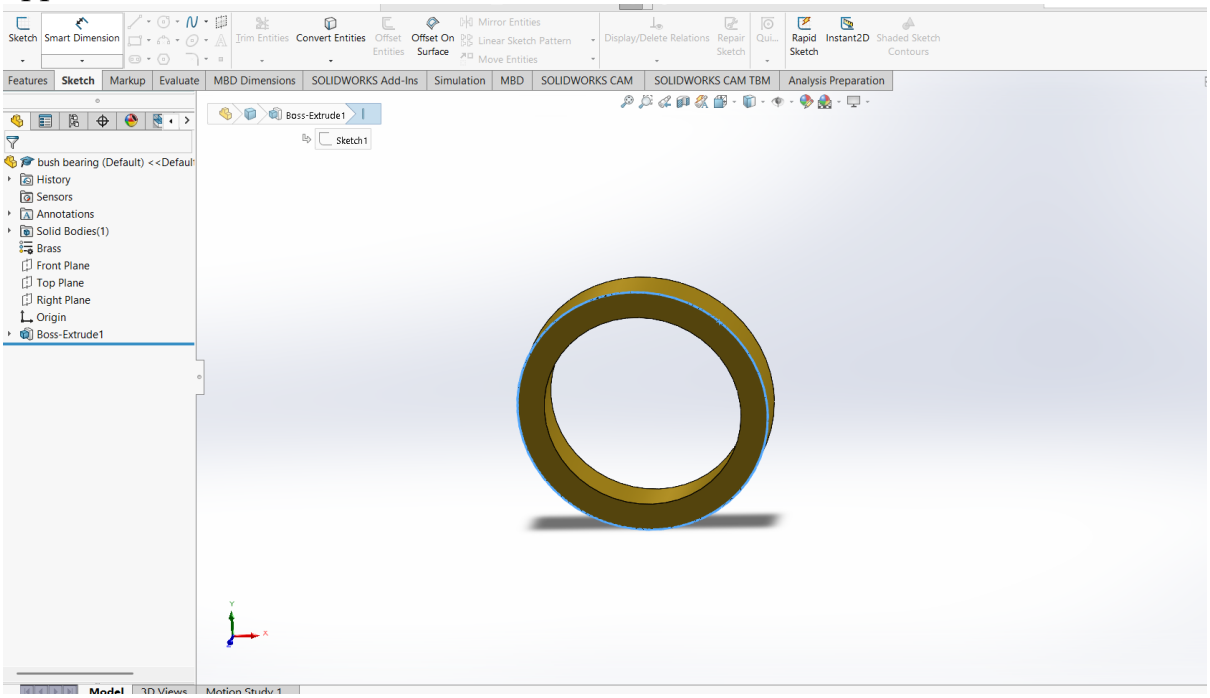
Ball bearings are an indispensable component in this model of the Spot Robot. They are used especially for reducing the friction between the rotating shaft and the upper limb. We use ball bearings instead of hydraulic and pneumatic bearings due to the simplicity of the design involved in ball bearings. Hydraulic and pneumatic bearings involve complex mechanisms which cannot be accommodated into the small legs of the Spot Robot. The ball bearings that I have used in my model are the SKF bearing_61804.



Ball Bearing

9) Bush bearings:

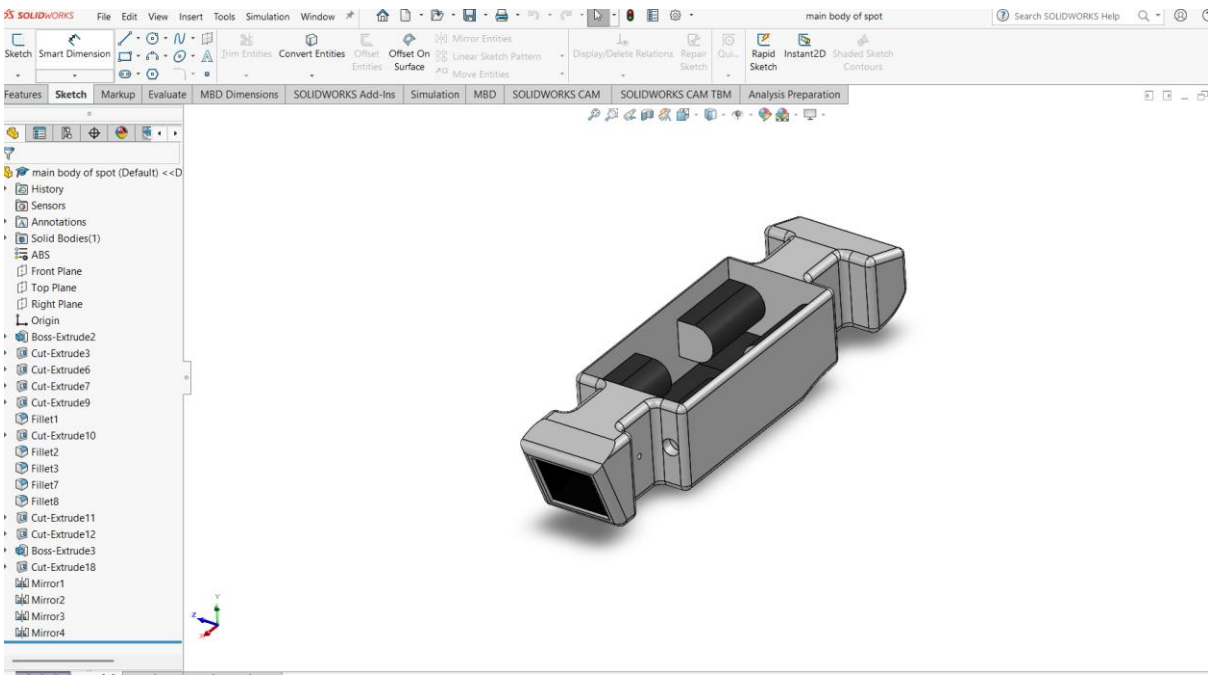
These are important for providing a cushioning layer between the upper limb and lower limb to prevent sparks caused due to friction that can prove to be detrimental to all the rotating components inside. The bush bearing is actually made of Brass which provide a low friction interface between the upper and lower limbs.



Bush Bearings

10) Main body of the Spot Robot:

This is the largest part that was designed using part modelling. The main body has 4 main motors embedded into it as shown. It was made by first making a 2-D sketch using basic sketch tools like straight line tool, spline, 3-point arc and the circular tools. Fillets of radius 12.50mm were applied at all major corners and a rectangular shaped hole which converges into the body was made at the front face of the bot. This is basically where the camera is installed. The material that is used for the main body was Acrylonitrile Butadiene Styrene. It is most suitable because it is fairly light, has good strength and can be used for housing electronics due to great heat resistance and electrical insulation. A hollow portion is made on the top only to show the motors and the space where the electronics can be accommodated. Holes are made on the 4 corners of the main body as shown in order to accommodate the counter sunk key mechanism that can be driven by the motors in the main body.



Main Body

MATHEMATICAL CALCULATIONS

Mass estimations and center of mass location:

The mass of the entire model of the bot was found using the mass properties tool in SolidWorks. Following list gives us the mass, volume, total surface area, position of center of mass and moments of intertia in tensor notation along different axes.

Mass = 24067.59 grams

Volume = 17355697.21 cubic millimeters

Surface area = 2389296.42 square millimeters

Center of mass: (millimeters)

X = -186.19

Y = 194.51

Z = 318.70

Principal axes of inertia and principal moments of inertia: (grams * square millimeters)

Taken at the center of mass.

Ix = (1.00, 0.04, 0.00) Px = 442247582.34

Iy = (0.00, -0.01, -1.00) Py = 2069430173.95

Iz = (-0.04, 1.00, -0.01) Pz = 2270778182.04

Moments of inertia: (grams * square millimeters)

Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)

Lxx = 445083760.45 Lxy = 71852995.73 Lxz = 3593868.17

Lyx = 71852995.73 Lyy = 2267933955.68 Lyz = 1955319.24

Lzx = 3593868.17 Lzy = 1955319.24 Lzz = 2069438222.20

Moments of inertia: (grams * square millimeters)

Taken at the output coordinate system. (Using positive tensor notation.)

Ixx = 3800285721.44 Ixy = -799788367.99 Ixz = -1424574743.46

Iyx = -799788367.99 Iyy = 5546893612.23 Iyz = 1493949955.88

Izx = -1424574743.46 Izy = 1493949955.88 Izz = 3814387378.13

ASSEMBLY MODELLING PROCESS

1)The main body of the Spot Robot has been chosen as the base part in this case, and then components/parts are inserted one by one and using a wide variety of mates the bot was assembled. Standard, advanced and mechanical mates were used in the assembly modelling process to complete the model of the bot.

2)Concentric mates were used to mate the shoulder shaft concentric to the holes on the 4 sides of the main body. The counter sunk key ws then inserted and was then mated with its hole concentric to the M5 tapped hole of the shoulder shaft.

3)This counter sunk key was then mated co-incident to the upper leg of the Spot Robot.

4)This process was repeated on the other 3 sides of the main body.

5)Now inside the upper leg of the Spot Robot we have one hub motor.

6) The screw and square shaped nut components are inserted into the assembly.

7)The screw is mated concentric to the round shaped top of the hub motor and is connected to this hub motor using a co-incident mate.

8) The square shaped nut also has threads on the centre hole, which is mated concentric to the screw.

9) The nut (as mentioned earlier) has an M5 tapped hole at the sides which is mated to the connecting link.

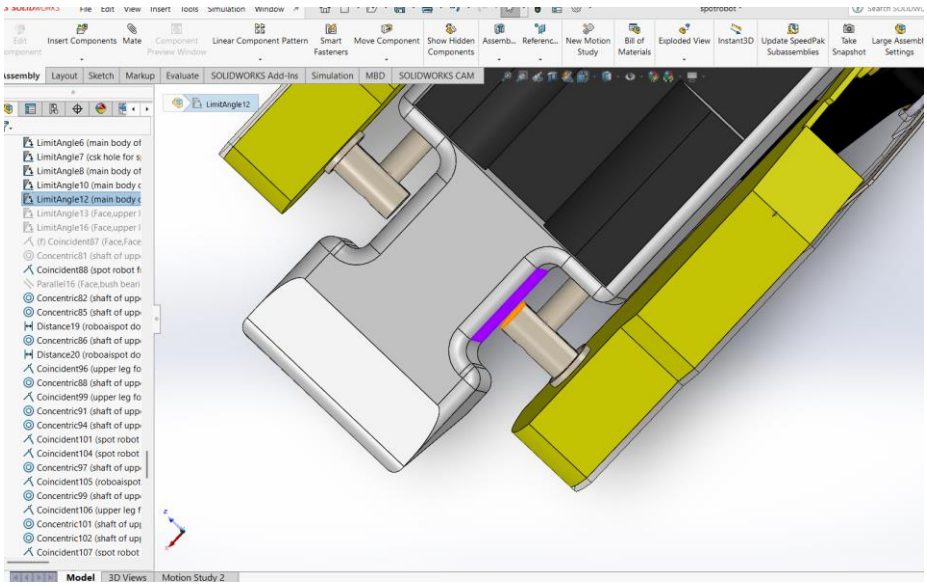
10) Then connecting link is in turn connected and mated concentric to the circular hole at the top of the lower limb with the help of a pin. Thus, establishing a connection between the upper and lower limbs.

11) In addition to the concentric mate we also use a mechanical screw mate to create a helical joint between the nut and the screw. The pitch value is set to 2mm/ revolution. Thus, when the screw connected to the motor rotates when the motor is on, the nut exhibits a linear reciprocating motion. In other words, this helical mate that we have created basically converts the rotary motion of the screw to linear reciprocating motion of the nut, which causes the lower limb to move.

11) Advanced mates have also been used in this assembly.

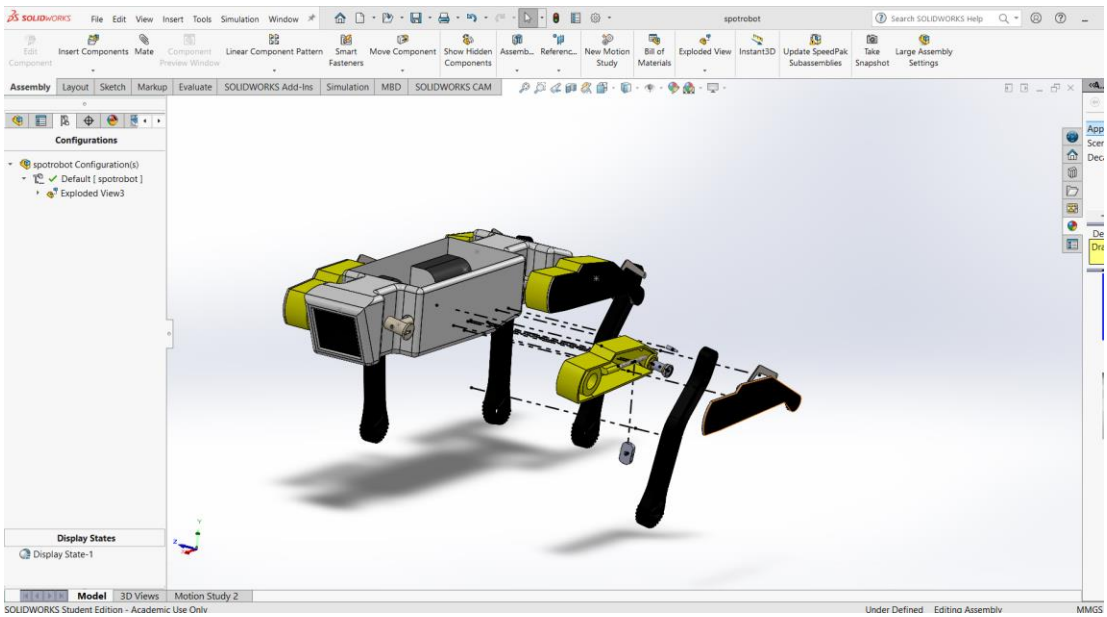
12) The limiting angle mate is one such advanced mate. This mate is normally used to define a specific angular range within which we want our part to rotate.

13) Similarly, the counter sunk key-shoulder shaft mechanism that connects the upper limb to the main body has also been restricted to a rotation of 10 degrees of either side with respect to the flat surface of the main body as shown here:

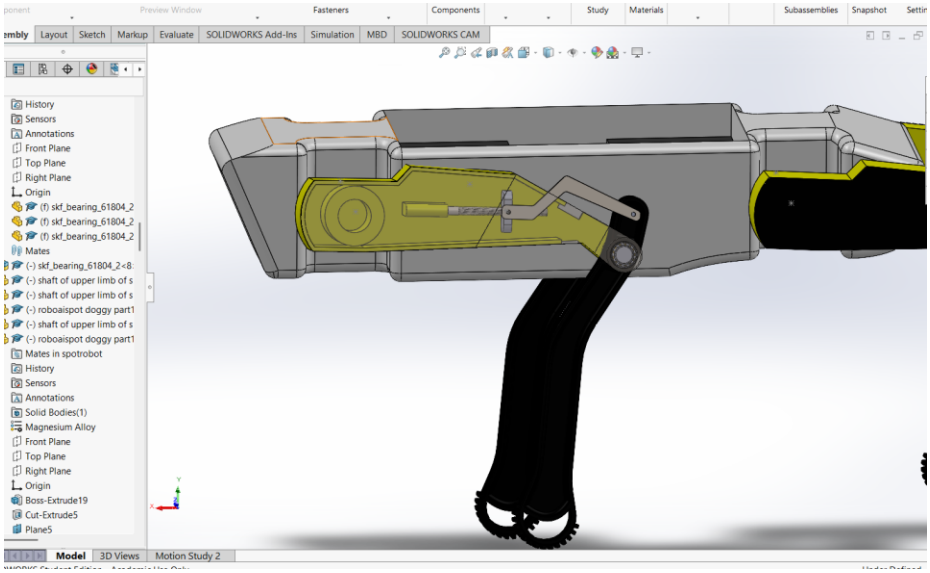


That was a brief description of the various mates that were used to bring this assembly together.

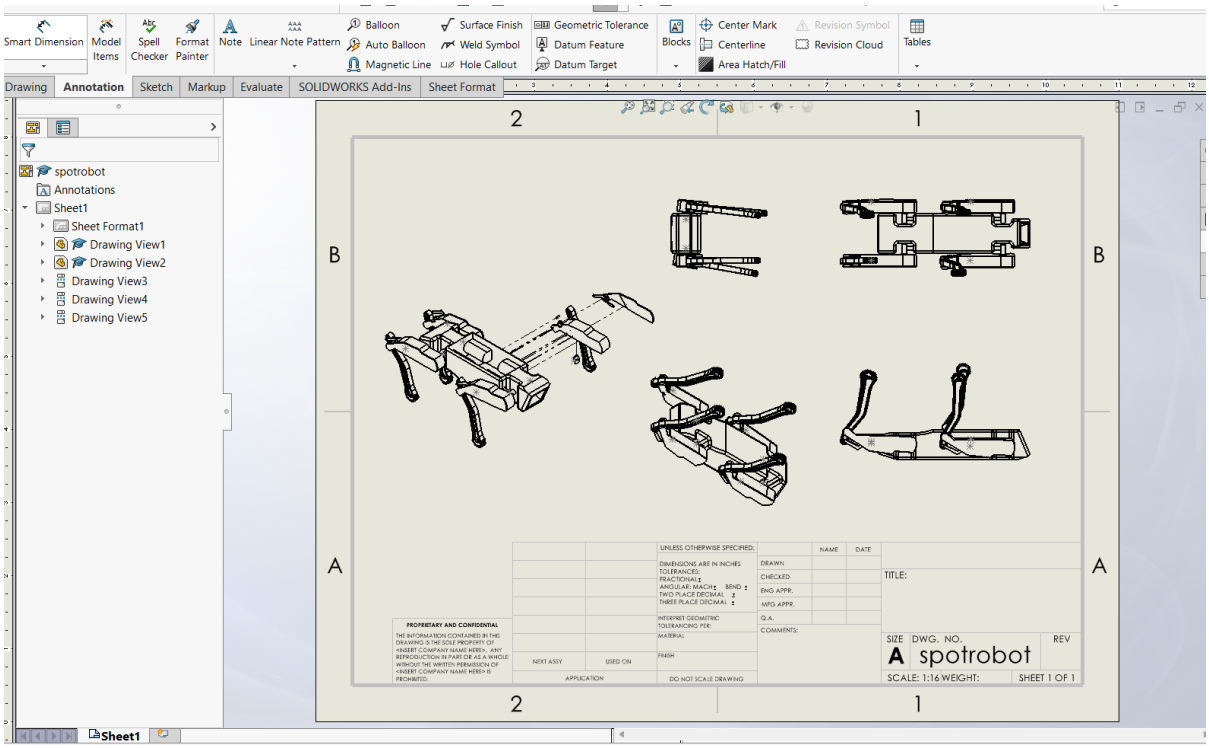
Exploded view of limbs of the model:



On making the cap covering the upper limb transparent we get the following image:



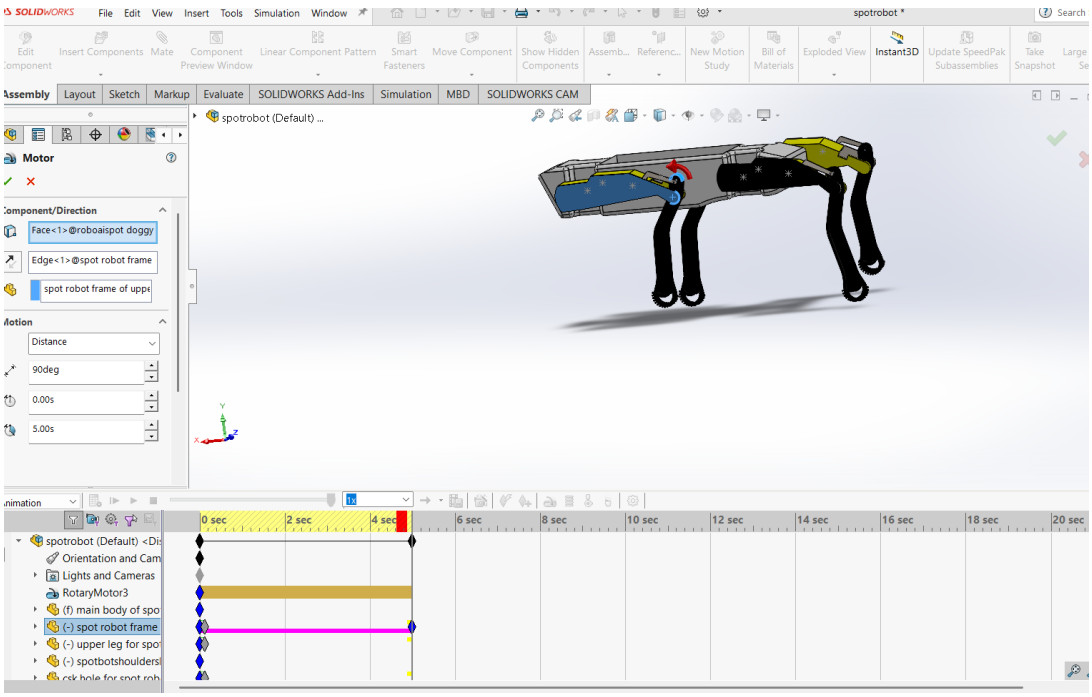
Exploded view in drawing format with top view, side views and front views:



SIMULATION OF THE MODEL

After completion of the assembling process, simulation of the movement of the limbs was done using the motion study tool in SolidWorks. We can carry out various simulations using various features of the motion study tab. One such feature is the motor feature which when applied on a component, simulates that particular component as if it were driven by a motor.

In this case the motoring action has been applied to the rounded edge of the lower limb so that the lower limb rotates as shown. The simulation time period has been set to 5 seconds and the direction too can be set while applying the motoring action.



CONCLUSION

Working on the 3D CAD model of a Spot-like robot has been an exciting and insightful journey. From conceptualizing the design to refining each component in SolidWorks, this project offered a hands-on experience in robotic modelling and structural planning. Every iteration helped improve the robot's balance, movement, and overall feasibility, bringing it one step closer to real-world application.

While this model is a solid starting point, there's still plenty of room for growth. Future steps could involve selecting the right motors, integrating control systems, and even building a working prototype. Enhancing stability, improving energy efficiency, and adding advanced sensors are all potential upgrades. This project not only deepened my understanding of robotics but also opened doors for further innovation in the field.